

# Musterlösung Serie 16

## DETERMINANT

1. Show that the determinant of a  $2 \times 2$  matrix is an alternating 2-linear function of rows and columns.

*Solution:*

Let  $K$  be a field, and let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(K).$$

Recall that

$$\det(A) = ad - bc.$$

We start by showing 2-linearity in the columns. Write the columns of  $A$  as

$$C_1 = \begin{pmatrix} a \\ c \end{pmatrix}, \quad C_2 = \begin{pmatrix} b \\ d \end{pmatrix}.$$

*Linearity in the first column.*

Fix  $C_2$  and suppose

$$C_1 = \alpha C'_1 + \beta C''_1, \quad \alpha, \beta \in K.$$

Write

$$C'_1 = \begin{pmatrix} a' \\ c' \end{pmatrix}, \quad C''_1 = \begin{pmatrix} a'' \\ c'' \end{pmatrix}.$$

Then

$$C_1 = \begin{pmatrix} \alpha a' + \beta a'' \\ \alpha c' + \beta c'' \end{pmatrix}.$$

Hence

$$\det(C_1, C_2) = (\alpha a' + \beta a'')d - b(\alpha c' + \beta c'').$$

Using distributivity and commutativity in  $K$ ,

$$\det(C_1, C_2) = \alpha(a'd - bc') + \beta(a''d - bc'') = \alpha \det(C'_1, C_2) + \beta \det(C''_1, C_2).$$

Thus the determinant is linear in the first column.

*Linearity in the second column.*

Fix  $C_1$  and let

$$C_2 = \alpha C'_2 + \beta C''_2, \quad \alpha, \beta \in K.$$

A similar computation gives

$$\det(C_1, C_2) = a(\alpha d' + \beta d'') - (\alpha b' + \beta b'')c = \alpha \det(C_1, C'_2) + \beta \det(C_1, C''_2).$$

Hence the determinant is linear in the second column.

Therefore  $\det$  is 2-linear in the columns.

Next we will show the alternating property in the columns. If  $C_1 = C_2$ , then  $a = b$  and  $c = d$ , so

$$\det(A) = ad - bc = ac - ca = 0,$$

Thus the determinant vanishes whenever the two columns are equal, and hence it is alternating in the columns.

Next we will turn to showing 2-linearity and the alternating property in the rows. Write the rows of  $A$  as

$$R_1 = (a \ b), \quad R_2 = (c \ d).$$

Then

$$\det(A) = ad - bc.$$

If

$$R_1 = \alpha R'_1 + \beta R''_1,$$

a direct substitution into the formula  $ad - bc$  shows that the determinant is linear in  $R_1$ , and similarly in  $R_2$ .

If  $R_1 = R_2$ , then  $a = c$  and  $b = d$ , so

$$\det(A) = ab - ba = 0.$$

Hence the determinant is alternating in the rows as well.

Therefore the determinant on  $2 \times 2$  matrices over  $K$  is 2-linear in both its columns and its rows, and it is alternating.

2. Let  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  be a  $2 \times 2$  matrix over a field  $K$ .

(a) Show that the matrix  $A$  is invertible if and only if  $ad - bc$  is not zero.

(b) Show that

$$A^2 = 0 \quad \text{if and only if} \quad \det(tI - A) = t^2$$

as polynomials in  $K[t]$ .

*Solution:*

(a) Suppose first that  $ad - bc \neq 0$ . Consider the matrix

$$B = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

A direct computation shows

$$AB = \frac{1}{ad - bc} \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix} = I,$$

and similarly  $BA = I$ . Hence  $A$  is invertible.

Conversely, suppose  $A$  is invertible. Then there exists  $B$  such that  $AB = I$ . Taking determinants gives

$$\det(A) \det(B) = \det(I) = 1.$$

Since  $\det(A) = ad - bc$ , this implies  $ad - bc \neq 0$ .

(b) Let  $K$  be a field and let  $A \in M_2(K)$ . Show that

$$A^2 = 0 \quad \text{if and only if} \quad \det(tI - A) = t^2$$

as polynomials in  $K[t]$ .

Let  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ . Then

$$\det(tI - A) = \det \begin{pmatrix} t - a & -b \\ -c & t - d \end{pmatrix} = (t - a)(t - d) - bc.$$

Expanding,

$$\det(tI - A) = t^2 - (a + d)t + (ad - bc) = t^2 - (\operatorname{tr} A)t + \det(A).$$

( $\Rightarrow$ ) Suppose  $A^2 = 0$ . Then

$$A^2 = \begin{pmatrix} a^2 + bc & ab + bd \\ ca + dc & cb + d^2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Hence

$$a^2 + bc = 0, \quad d^2 + bc = 0, \quad b(a + d) = 0, \quad c(a + d) = 0.$$

Subtracting the first two equations gives

$$a^2 - d^2 = (a - d)(a + d) = 0.$$

If  $a + d \neq 0$ , then from  $b(a + d) = 0$  and  $c(a + d) = 0$  we obtain  $b = c = 0$ , and then  $a^2 = 0$  and  $d^2 = 0$ , hence  $a = d = 0$ , so  $a + d = 0$ , a contradiction. Therefore

$$a + d = 0.$$

Thus  $\text{tr } A = 0$ .

From  $a^2 + bc = 0$  and  $d = -a$ , we obtain

$$\det(A) = ad - bc = a(-a) - (-a^2) = -a^2 + a^2 = 0.$$

Therefore

$$\det(tI - A) = t^2 - (\text{tr } A)t + \det(A) = t^2.$$

( $\Leftarrow$ ) Conversely, suppose

$$\det(tI - A) = t^2.$$

Comparing with

$$\det(tI - A) = t^2 - (\text{tr } A)t + \det(A),$$

we obtain

$$\text{tr } A = a + d = 0, \quad \det(A) = ad - bc = 0.$$

Hence  $a(-a) - bc = 0$  and similarly  $(-d)d - bc = 0$ , and clearly  $b(a + d) = 0$  and  $c(a + d) = 0$ . Thus

$$A^2 = \begin{pmatrix} a^2 + bc & ab + bd \\ ca + dc & cb + d^2 \end{pmatrix} = 0.$$

Hence

$$A^2 = 0 \quad \text{if and only if} \quad \det(tI - A) = t^2.$$

3. Each of the following expressions defines a function  $D$  on the set of  $3 \times 3$  matrices over the field of real numbers. In which of these cases is  $D$  a 3-linear function?

- (a)  $D_1(A) = A_{11} + A_{22} + A_{33}$ ;
- (b)  $D_2(A) = A_{11}^2 + 3A_{11}A_{22}$ ;
- (c)  $D_3(A) = A_{11}A_{12}A_{33}$ ;
- (d)  $D_4(A) = A_{13}A_{22}A_{32} + 5A_{12}A_{22}A_{32}$ ;
- (e)  $D_5(A) = 0$ ;
- (f)  $D_6(A) = 1$ .

*Solution:* Let

$$B = \begin{pmatrix} 1 & 1 & \\ & 1 & \\ & & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 2 & \\ & 1 & \\ & & 1 \end{pmatrix}.$$

These matrices will give us counterexamples for (a), (b), (c) and (f). Indeed, for (a),

$$D_1(C) = 4 \neq 6 = 2(1 + 1 + 1) = 2D_1(B).$$

For (b),

$$D_2(C) = 10 \neq 8 = 2(1 + 3 \cdot 1 \cdot 1) = 2D_2(B).$$

For (c),

$$D_3(C) = 4 \neq 2 \cdot 1 \cdot 1 \cdot 1 = 2D_3(B).$$

Finally, for (f),

$$D_6(C) = 1 \neq 2 = 2D_6(B).$$

We deduce that in these 4 cases, the map is **not** 3-linear.

We now denote the rows of an arbitrary matrix  $A$  by  $R_i$ ,  $i = 1, 2, 3$  and view  $D(A) = D(R_1, R_2, R_3)$  as a function on the rows of  $3 \times 3$  real matrices. Let  $\alpha = (\alpha_i)_{i=1}^n$  be a row vector and let  $\lambda \in \mathbb{R}$ . Then, in the case of (d),

$$\begin{aligned} D_4(\lambda R_1 + \alpha, R_2, R_3) &= (\lambda A_{13} + \alpha_3)A_{22}A_{32} + 5(\lambda A_{12} + \alpha_2)A_{22}A_{32} \\ &= \lambda(A_{13}A_{22}A_{32} + 5A_{12}A_{22}A_{32}) + (\alpha_3A_{22}A_{32} + 5\alpha_2A_{22}A_{32}) \\ &= \lambda D_4(R_1, R_2, R_3) + D_4(\alpha, R_2, R_3). \end{aligned}$$

This shows that  $D_4$  is linear in the first row and similar computations prove that  $D_4$  is linear in the second and third row, hence that it is 3-linear.

Finally, for (e),

$$D_5(\lambda R_1 + \alpha, R_2, R_3) = 0 = \lambda \cdot 0 + 0 = \lambda D_5(R_1, R_2, R_3) + D_5(\alpha, R_2, R_3)$$

and similarly for rows 2 and 3. This shows that  $D_5$  is 3-linear.

4. Let  $K$  be a subfield of the complex numbers and  $n$  a positive integer. Let  $j_1, \dots, j_n$  and  $k_1, \dots, k_n$  be positive integers not exceeding  $n$ . For an  $n \times n$  matrix  $A$  over  $K$  define

$$D(A) = A(j_1, k_1)A(j_2, k_2) \cdots A(j_n, k_n).$$

Prove that  $D$  is  $n$ -linear if and only if the integers  $j_1, \dots, j_n$  are distinct.

*Solution:* We first assume that  $j_r = j_s$  for some  $r \neq s$  and show that  $D$  is not  $n$ -linear in this case. Let  $B$  be the matrix obtained by multiplying the  $j_r$ -th row of  $A$  by  $\lambda \in K$ . Then,

$$\begin{aligned} D(B) &= \prod_{\substack{i=1 \\ j_i \neq j_r}}^n A(j_i, k_i) \cdot \prod_{\substack{k=1 \\ j_k = j_r}}^n (\lambda A(j_r, k_r)) \\ &= \lambda^n D(A), \end{aligned}$$

where  $m \geq 2$  is the number of indices  $s$  for which  $j_s = j_r$ . Since in general  $\lambda^m D(A) \neq \lambda D(A)$ , e.g. for  $\lambda = 2$  and non-zero matrix entries, this shows that  $D$  is not  $n$ -linear.

*Note.* In the case of  $K := \mathbb{F}_2$ , the field with 2 elements, the scalars are the additive identity  $0_K$  and the multiplicative identity  $1_K$ . It follows from the field axioms (check it!) that  $\forall m \geq 1 : 0_K^m = 0_K$ , and  $\forall m \geq 0 : 1_K^m = 1_K$ . Hence such a map  $D$  would be  $n$ -linear in this case. However, since  $\mathbb{F}_2$  is not a subfield of  $\mathbb{C}$  (to see why, try to define an injective field homomorphism from  $\mathbb{F}_2$  into  $\mathbb{C}$ ), this cannot happen here.

Let us now assume that the  $j_i$ 's are pairwise distinct. Let  $\lambda \in K$ , let  $1 \leq r \leq n$ , let  $\{\alpha_i\}_{i=1}^n \subset K$ , and consider  $B$  to be the matrix obtained by multiplying the  $j_r$ -th row of  $A$  by  $\lambda$  then adding to it the row-vector  $(\alpha_i)_{i=1}^n$ . We compute that

$$\begin{aligned} D(B) &= \prod_{\substack{i=1 \\ i \neq r}}^n A(j_i, k_i) \cdot (\lambda A(j_r, k_r) + \alpha_{k_r}) \\ &= \lambda D(A) + D(C), \end{aligned}$$

where  $C$  is the matrix obtained by replacing the  $j_r$ -th row of  $A$  by  $(\alpha_i)_{i=1}^n$ . This shows that  $D$  is  $n$ -linear.

5. Let  $K$  be a field. Define a function  $D$  on  $3 \times 3$  matrices over  $K$  by the rule

$$D(A) = A_{11} \det \begin{bmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{bmatrix} - A_{12} \det \begin{bmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{bmatrix} + A_{13} \det \begin{bmatrix} A_{21} & A_{22} \\ A_{31} & A_{32} \end{bmatrix}.$$

Show that  $D$  is alternating and 3-linear as a function of the columns of  $A$ .

*Solution:* Let  $A = (A_{ij}) \in M_3(K)$ . Let the columns of  $A$  be denoted by  $C_1, C_2, C_3 \in K^3$ . We prove that  $D$  is 3-linear and alternating in the columns.

We start by showing it is 3-linear. Each  $2 \times 2$  determinant appearing in the definition is bilinear in its two columns. Since multiplication in  $K$  is commutative, each term in the definition of  $D$  is a product of an entry from one column with a bilinear expression in the remaining two columns.

*Linearity in the first column.*

Fix  $C_2, C_3$ . Write

$$C_1 = \alpha C'_1 + \beta C''_1, \quad \alpha, \beta \in K.$$

Then each entry  $A_{i1}$  is linear in  $C_1$ , while the  $2 \times 2$  minors do not involve entries from  $C_1$ . Hence

$$D(\alpha C'_1 + \beta C''_1, C_2, C_3) = \alpha D(C'_1, C_2, C_3) + \beta D(C''_1, C_2, C_3).$$

*Linearity in the second column.*

Fix  $C_1, C_3$ . Each  $2 \times 2$  determinant involving the second column is linear in that column, and the coefficients multiplying these determinants do not depend on  $C_2$ . Therefore

$$D(C_1, \alpha C'_2 + \beta C''_2, C_3) = \alpha D(C_1, C'_2, C_3) + \beta D(C_1, C''_2, C_3).$$

*Linearity in the third column.*

The same argument applies, since the minors are linear in the third column as well. Thus

$$D(C_1, C_2, \alpha C'_3 + \beta C''_3) = \alpha D(C_1, C_2, C'_3) + \beta D(C_1, C_2, C''_3).$$

Hence  $D$  is linear in each column separately; therefore it is 3-linear.

To show  $D$  is alternating, we show that if two columns are equal, then  $D(A) = 0$ .

*Case  $C_1 = C_2$ .*

Then  $A_{i1} = A_{i2}$  for all  $i$ . Substituting into the definition gives

$$D(A) = A_{11} \begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} - A_{11} \begin{vmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{vmatrix} + A_{13} \begin{vmatrix} A_{21} & A_{22} \\ A_{31} & A_{32} \end{vmatrix}.$$

Since  $A_{21} = A_{22}$  and  $A_{31} = A_{32}$ , the third  $2 \times 2$  determinant vanishes. Expanding the first two determinants and using the equalities  $A_{21} = A_{22}$  and  $A_{31} = A_{32}$ , one verifies that the remaining terms cancel pairwise, yielding  $D(A) = 0$ .

*Case  $C_1 = C_3$  or  $C_2 = C_3$ .*

The same computation shows that the corresponding pairs of terms cancel and the remaining minor vanishes. Hence  $D(A) = 0$  whenever two columns coincide.

Since  $D$  is 3-linear and vanishes whenever two columns are equal, it is an alternating 3-linear function of the columns of  $A$ .